

$$C = B \log_2 (1 + \text{SNR})$$

Shannon: what is possible

$$y = \int r(t) s^*(t) dt$$

matched filter: correlate with
what you expect

$$P_r = P_t + G_t + G_r - L$$

the link budget: never repealed

$$X(f) = \int x(t) e^{-j2\pi ft} dt$$

Fourier: time/frequency, delay/
Doppler, aperture/angle

$$2BT, \pi A/\lambda^2$$

degrees of freedom: count them

$$-174 \text{ dBm/Hz}$$

noise is the floor, and it is kT